

Multitask Learning for Tree Instance Segmentation and Diameter Estimation from RGB-D Imagery in Agroforestry and Rainforest Environments

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Extended reading summary of the full research paper, styled to match the project poster. Every main-body table and figure is included; the appendices, the CRediT and funding declarations, and references beyond those cited here are left out. The full 28-page paper covers the same work with the complete methodology and appendices.

Can one photo, taken by a field crew standing under the canopy, replace a diameter tape and a species guidebook?

62.29%

TEST AP₅₀

7.62 cm

TEST DBH ERROR

+12.92 pts

FROM ADDING DEPTH

Abstract. Tree inventory is a pivotal process in forest management, and it relies on tree identification and parameters such as diameter at breast height (DBH) and height. Small errors in measurement or identification can lead to miscalculation of tree volume and carbon accounting. In Indonesia, inventory is mostly carried out manually due to the high cost of terrestrial laser scanning and limited imagery of local tree species. This research trains MaskDINO, a computer vision model, on four-channel images, leveraging the architecture of the vision foundation model FocalNet-L as its backbone. The model is trained using SPREAD, a synthetic dataset. A segmentation head identifies tree species, while a second head estimates DBH from backbone features at the image row corresponding to breast height, so a single model identifies species and estimates DBH from ground-level RGB-D images captured under the canopy. The final model, trained with diameter supervision across eleven species, achieves a test AP₅₀ of 62.29 %, with no measurable cost to segmentation compared with an otherwise identical segmentation-only model. Evaluated end-to-end on its own detections rather than ground-truth annotations, it estimates diameter for 27.1 % of eligible trees, with a DBH root-mean-square error of 7.62 cm. In a controlled comparison where both models are trained from scratch and differ only in channel count, adding depth increases AP₅₀ by 12.92 percentage points. Detection recall falls with camera distance, from 82.4 % for trees within 5 m to 34.5 % beyond 30 m, while species classification and relative diameter error remain largely stable for detected trees.

1 Introduction

Indonesia's tropical and plantation forests are exceptionally biodiverse and carbon-dense, both of which make them central to global climate regulation (Baccini et al., 2017). Managing these forests rests on forest inventory, which turns standing vegetation into numbers that policy can act on. Carbon stock estimates, biodiversity indices, and timber yield models are all built from per-tree records, of which species, trunk diameter, and height matter most, and this study addresses the first two. Diameter at breast height (DBH) enters allometric biomass equations directly (Manuri et al., 2016), so error in DBH propagates into every downstream carbon figure (Baccini et al., 2017). Across Indonesian forests and agroforestry systems, these records are still collected by hand with diameter tapes. Its cost lies in access, repetition, and the skilled

labour needed to identify species in the field, and it scales with the number of trees rather than the area surveyed.

Two technologies have been proposed to reduce that cost. Terrestrial laser scanning (Liang et al., 2018) recovers trunk geometry with millimetre precision but requires equipment whose price places it outside routine operational inventory in much of the region. Photogrammetric and image-based approaches use commodity cameras, but supervised learning needs annotated imagery, and per-instance annotation of tropical species under closed canopy barely exists. That deep learning can segment trees is well established, at least in aerial and plantation settings. Two harder questions remain. Can it do so from the viewpoint a field crew occupies beneath the canopy? Can it deliver the second measurement that inventory needs, rather than delineation alone?

Prior approaches differ mainly in acquisition protocol. Close-range work captures one trunk from a metre or two, with a smartphone photograph (Holcomb et al., 2023; Wu et al., 2019; Xiang et al., 2025), a coarse depth map (Feng et al., 2024), or monocular SLAM (Fan et al., 2018; Su et al., 2024), and reaches errors of a few centimetres. That accuracy is real, but the operator chooses the viewpoint and isolates the trunk before the model runs. Accumulating many views per trunk, from an under-canopy UAV (García-Pascual et al., 2025) or a 360-degree camera carried through the stand (He et al., 2025), improves accuracy the same way, by spending more effort per tree. Work on whole scenes instead tends to deliver one attribute rather than two, whether detection and species without diameter (Duclos et al., 2026), trunk detection in urban scenes (Li et al., 2023b), diameter in planted stands where trunks are separable (Lu et al., 2026), or aerial crown delineation from a viewpoint where trunks are not visible (Ball et al., 2023). The closest precedent is Grondin et al. (2022), which detects multiple trees and estimates diameter from ground-level synthetic renders without resolving species, the axis this study adds.

The under-canopy viewpoint complicates both instance segmentation and diameter estimation. Aerial and unmanned-aerial imagery observes crowns from above, where individual trees are largely separable and occlusion is limited to overlapping crown margins. Airborne segmentation procedures recover more than 90 % of overstory trees but barely 60 % of understory trees, because the upper canopy occludes what lies beneath (Hamraz et al., 2017). Under-canopy imagery inverts that problem rather than removing it: trees occlude one another heavily and apparent size varies by more than an order of magnitude with distance to the camera. The trunk at breast height, the attribute inventory most needs, is frequently hidden behind foreground vegetation. Tropical stems compound this, since large trees are often buttressed and markedly non-circular, which the geometric fitting traditionally used for diameter estimation cannot accommodate (Nölke et al., 2015). Depth sensing addresses this geometry directly, and without a distance to the trunk pixel width cannot be converted to a physical width at all.

This study addresses three questions. First, how much does an explicit depth channel contribute to instance segmentation under dense canopy, measured under controls that isolate the channel from the initialisation strategy it is usually confounded with. Second, can a single network produce species-resolved instance masks and trunk diameter together, and what does that coupling cost relative to training the two objectives separately. Third, once performance is stratified by distance from the camera, where do the errors lie.

The answer is a multitask RGB-D network trained on SPREAD (Feng et al., 2025), which renders ground-level forest scenes in Unreal Engine 5, spanning Indonesian agroforestry species and rainforest vegetation in the same partition. This study leverages the architecture of FocalNet-L (Yang et al., 2022), a vision foundation model, as the backbone of MaskDINO (Li et al., 2023a). Its input convolution is widened to accept depth as a fourth channel, and the model is trained from scratch on this four-channel input. A second head regresses diameter from backbone features sampled at the image row that corresponds to breast height. Segmentation covers all thirteen species in the dataset, while diameter is supervised on the eleven for which ground-truth DBH is recorded.

The contributions are as follows. A controlled measurement of the depth channel, reporting pairs in which both arms are trained from scratch and only the channel count differs, removing the usual confound with pretrained initialisation. A multitask architecture that couples segmentation and diameter through shared features read at a geometrically derived breast-height row,

with a stop-gradient that keeps the coupling one-directional. A distance-stratified evaluation, made possible because depth is an input and every ground-truth instance therefore carries a physical distance independent of model weights, separating three abilities that aggregate metrics conflate: finding a tree, naming it, and measuring it. And an explicit account of what does not work: a geometric diameter baseline that fails at this resolution, and a loss weighting schedule that degrades segmentation without improving diameter.

Together, these contributions point toward a low-cost alternative to terrestrial laser scanning. A single commodity RGB-D camera, processed by one network, could let resource-limited forestry agencies and smallholder programs track stand structure and carbon stock without specialised equipment or field taxonomists, once the sim-to-real gap discussed in Section 4.4 is closed.

2 Materials and methods

2.1 Study data

This study uses SPREAD (Feng et al., 2025), a synthetic dataset rendered in Unreal Engine 5 that simulates ground-level under-canopy views of forest and plantation scenes. Each sample provides a colour image, a per-pixel depth map in metres, per-instance segmentation masks, a species label, and, for a subset of instances, a recorded DBH. The virtual camera stands about 2 m above ground and looks laterally into the stand, reproducing the occlusion and distance-dependent scale variation of a field crew’s viewpoint. All imagery is rendered at 480×270 pixels with a vertical focal length of 240 px, a fixed property of the released data that places a floor on attainable diameter accuracy (Section 3.6).

Thirteen species are retained, selected for their occurrence in Indonesian agroforestry and rain-forest stands, spanning a plantation domain of orchard and fruit species and a rainforest domain of figs, palms, understorey vines, and ferns. Table 1 reports the split sizes. The partition is made at the level of the rendered scene rather than the individual image, because one scene yields several images and an image-level split would place the same physical tree on both sides of it. An annotation audit preceded all experiments; repairing 416 masks, 0.8 % of instances, improved a pretrained RGB configuration by 1.7 points of AP_{50} while leaving an RGB-D configuration trained from scratch unchanged within noise.

Table 1: Dataset partitions. Instance counts are annotated tree instances; not all carry a recorded DBH.

Partition	Images	Instances
Train	3968	37 105
Validation	852	7870
Test	843	7925

Diameter supervision covers eleven of the thirteen species; the two excluded carry no recorded DBH for more than half of their instances. Segmentation continues to use all thirteen classes, so the restriction changes only which instances contribute to the diameter loss. The eleven species admit 1790 test instances with a well-posed breast-height geometry inside 20 m. Four of the eleven are small fruit trees whose form makes a single diameter at 1.3 m an imperfect description, discussed in Section 4.4.

2.2 Architecture

This study builds on MaskDINO (Li et al., 2023a), a query-based instance segmentation architecture that predicts masks, boxes, and classes jointly, with a FocalNet-L (Yang et al., 2022) backbone. Fig. 1 shows the configuration used. The backbone emits four feature levels, res_2 at one quarter of the input resolution through res_5 at one thirty-second. For FocalNet-L, res_2 carries 192 channels and measures 120×68 pixels at the native input size, the finest level available to any head that needs to resolve a trunk a few pixels wide. A pixel decoder projects all four levels to a common width of 256 and also produces the proposals that initialise the object

queries, which the transformer decoder then refines with deep supervision. The architecture is adopted unmodified except at the input stem, whose 7×7 convolution is widened from three to four channels to accept RGB-D input, and the classification head, which predicts 13 classes to match this study's species set.

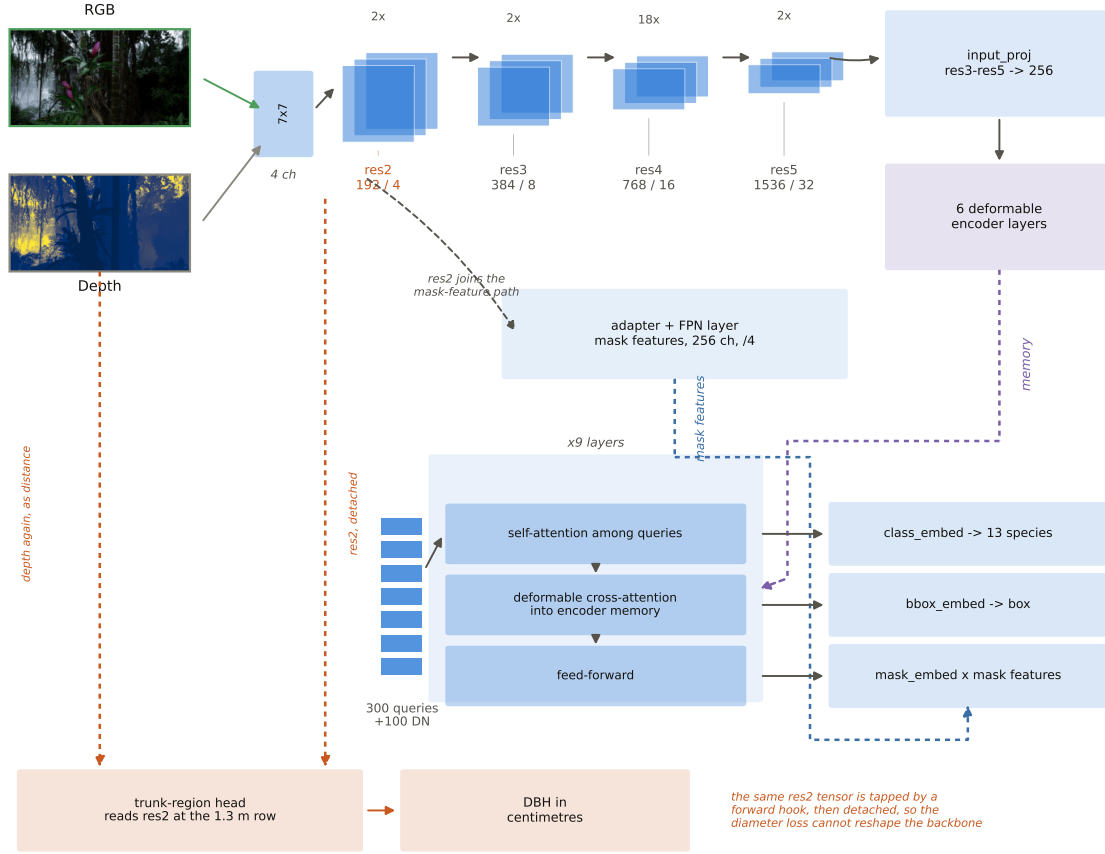


Figure 1: Architecture of the final model. RGB and depth form one four-channel input to a single FocalNet-L backbone. The segmentation decoder and the trunk-region diameter head both read the same `res2` feature; the diameter head reads it through a stop-gradient, so its loss never reshapes the backbone. Depth enters twice, once as an input channel and once as the geometry that converts pixel width into physical width.

Backbone selection proceeded by ablation over ResNet-50, Swin-T, Swin-B, Swin-L, FocalNet-B, and FocalNet-L, together with a Mask R-CNN reference to establish that the query-based architecture is itself responsible for a large part of the performance. Because four-channel input cannot inherit three-channel pretrained weights, RGB-D configurations are trained from scratch, which here requires full-precision arithmetic since mixed precision diverges within the first few hundred iterations. The ablations use a 75 000-iteration protocol with the diameter objective disabled, so that a full matrix of runs is affordable; the final model instead runs for 155 000 iterations with both objectives active, so ablation values sit below what the same configuration reaches at the final model's budget, an offset Section 3.3 quantifies.

2.3 Trunk-region diameter head

Diameter is predicted by an auxiliary head that reads a backbone feature directly rather than the decoder queries. The head taps `res2` through a forward hook on the backbone, so it consumes the same tensor the segmentation branch consumes rather than a recomputed copy. DBH is defined at a specific height on the trunk, so a head that pools features over an entire crown discards the localisation the measurement requires.

The head takes an instance mask (ground truth during training, the model's own prediction at inference) and the depth map, and locates the image row whose back-projected world height is closest to 1.3 m,

$$h(r) = h_{\text{cam}} - \frac{(r - c_y) d(r)}{f_y}, \quad (1)$$

where r indexes image rows, $d(r)$ is the median depth along the masked part of that row, c_y and f_y are the principal point and focal length in pixels, and h_{cam} is the camera height. Among the columns of the selected row the columns closest to the camera are retained, which isolates the trunk from foreground clutter sharing the same row, giving d_{trunk} as their mean depth and w_{px} as their count. This yields a purely geometric diameter estimate,

$$\hat{D}_{\text{geom}} = \frac{w_{\text{px}} d_{\text{trunk}}}{f_x}, \quad (2)$$

reported as a baseline and also supplied to the head as an explicit input feature. Fig. 2 traces these steps on one instance.

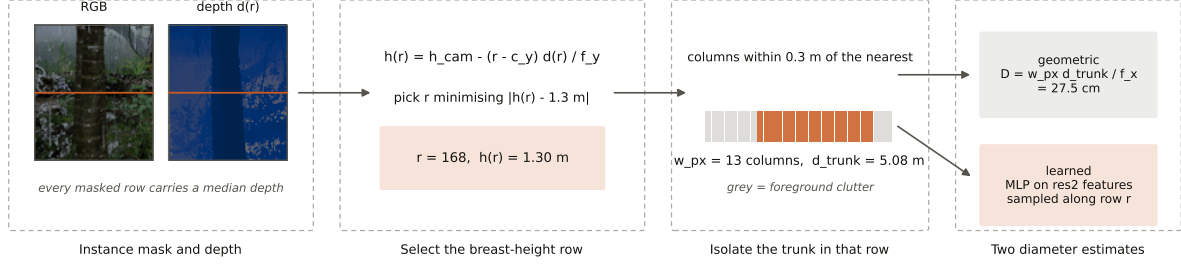


Figure 2: The steps above, traced on one validation instance, a BangaloPalm at 5.08 m. The selected row is 168, back-projected to 1.30 m, leaving 13 trunk columns and a geometric estimate of 27.5 cm. The learned head reads `res2` features along the same row, so the two estimates are not alternatives; the geometric value is itself an input to the head.

The head itself concatenates three sources: backbone `res2` features sampled from a strip of rows centred on the breast-height row, the geometric estimate together with d_{trunk} , and a species one-hot vector. A small multilayer perceptron regresses the diameter in per-species log space, which prevents the species with the largest stems from dominating the objective. Instances contribute to the diameter loss only if the geometry is well posed: at least five trunk columns, a back-projected height within 0.5 m of breast height, and a trunk distance inside the training range.

2.4 Multitask training

Both objectives share one backbone. A forward hook captures the `res2` feature map once per iteration, so the segmentation branch and the diameter head read identical features, but the two gradients are not treated alike: the captured feature map is detached before it enters the diameter head, so the head is fitted to the representation that the segmentation objective produces without contributing any gradient of its own to the backbone. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_{\text{DBH}} \mathcal{L}_{\text{DBH}}, \quad (3)$$

where $\mathcal{L}_{\text{seg}}$ is the standard MaskDINO combination of classification, mask, dice, box, and generalised IoU terms applied across all decoder layers and denoising queries, and $\mathcal{L}_{\text{DBH}}$ is a single scalar regression term, so nominally equal weights do not produce equal influence.

The final model is trained from scratch for 155 000 iterations at batch size four with AdamW, base learning rate 1×10^{-4} , weight decay 0.05, and step decay at iterations 124 000 and 144 667. The diameter term carries $\lambda_{\text{DBH}} = 5$, held constant for the whole schedule; Section 3.4 reports what happens when it is raised instead. The diameter loss admits instances out to 30 m, covering 99.0 % of the depth distribution; the tighter 20 m bound that appears in the reported diameter numbers is a convention of the evaluation, not of training.

2.5 Evaluation protocol

Segmentation is evaluated with COCO instance metrics; AP_{50} , the percentage of predicted masks whose overlap with a true tree reaches at least 50 % IoU, averaged across classes and confidence thresholds, is the standard detection-accuracy score used throughout. Checkpoints

are selected on the validation split and all reported results are measured on the held-out test split.

Diameter is evaluated end-to-end: the head reads the model’s own predicted mask and predicted species, for instances the model detects and for which the predicted mask yields a valid breast-height geometry, rather than the ground truth. A predicted instance is matched to a ground-truth one at $\text{IoU} \geq 0.5$, greedy over prediction score, and both must then pass the geometric filters above before the pair is scored. This is the condition a deployed system experiences. Where two configurations are compared directly, the comparison is restricted to instances for which both yield a diameter, so a difference in which trees survive detection cannot by itself produce a difference in the reported error. Root-mean-square error is the primary metric, supported by mean absolute error, bias, and the mean of per-species R^2 , since the aggregate R^2 is inflated by one abundant species.

Distance-stratified analysis uses d_{trunk} from Eq. 1, which depends only on the ground-truth mask and the depth map, not on model weights. Identical instances therefore populate identical bins across models, and any difference between models within a bin is attributable to the models alone.

3 Results

3.1 The multitask model

The final model trains segmentation and diameter together from scratch for 155 000 iterations, with diameter supervised on eleven species. Evaluated end-to-end, using the masks and species the model predicts for its own detections, it returns a diameter for 27.1 % of eligible trees in the test split (1690 of 6233). On those trees it reaches 7.62 cm RMSE, 4.89 cm MAE, a bias of -1.28 cm, and a mean per-species R^2 of 11.88 % (8 species with $N \geq 30$). Segmentation is 62.29 % test AP_{50} .

Table 2: FocalNet-L, FocalNet-B, and Swin-L at the final model’s own protocol, 155 000 iterations with the same stop-gradient and diameter objective. Diameter metrics are end-to-end on the 1391 test trees all three backbones resolve on their own.

Backbone	AP_{50} (%)	Recall (%)	RMSE (cm)	MAE (cm)	Bias (cm)	Per-sp. R^2 (%)
Swin-L	56.55	55.78	7.29	4.86	0.26	15.15
FocalNet-B	63.09	63.09	7.74	4.97	-1.64	9.03
FocalNet-L	62.29	61.72	7.40	4.79	-1.31	12.96

FocalNet-L was chosen over two other backbones carried to the same multitask protocol (Table 2). FocalNet-B reaches a marginally higher AP_{50} , within the run-to-run spread of Section 3.4, but FocalNet-L reaches a lower RMSE and a higher mean per-species R^2 , so FocalNet-L is used here because segmentation ties between the two while diameter does not.

Attaching the diameter objective is close to free on the segmentation side. Over the full schedule the multitask and segmentation-only training trajectories stay within run-to-run variation of each other, so the second output is obtained without measurably degrading the first.

3.2 Contribution of the depth channel

Comparisons of RGB against RGB-D are commonly confounded in the literature: a four-channel stem cannot inherit three-channel pretrained weights, so it is usually trained from scratch while its RGB counterpart starts from ImageNet initialisation. Holding the channel count fixed for FocalNet-L, pretraining raises test AP_{50} by 4.29 points on RGB but costs 7.73 points on RGB-D once the three-channel stem is naively inflated to four, since the ImageNet features that help a three-channel input work against a four-channel one. A comparison that pretrains both arms would therefore report an apparent depth advantage of only 0.91 points. Only when pretraining is removed from both arms does the 12.92-point advantage associated with depth become visible.

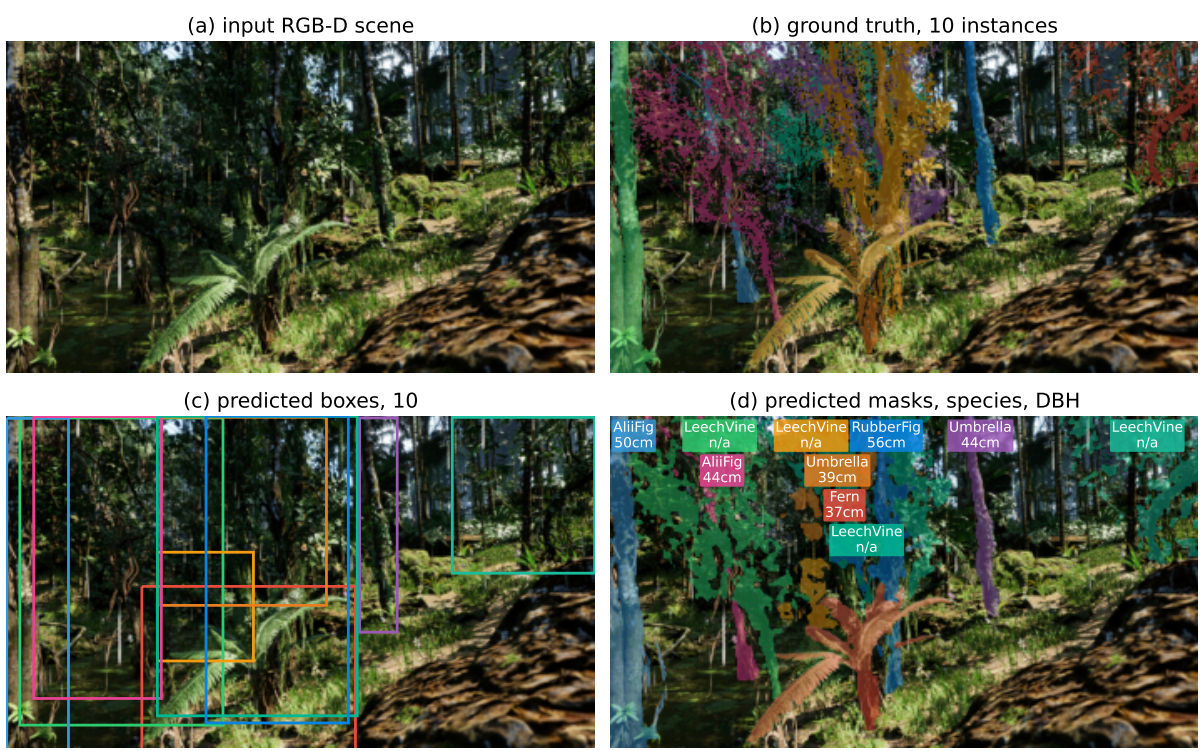


Figure 3: One rainforest scene from the test split, in four panels: the input RGB-D, the 10 annotated instances, predicted boxes, and predicted masks, all at score threshold 0.30 and from the same forward pass, with predicted species and diameter. Labels read *n/a* where the head found no valid breast-height row, typically an occluded trunk, or where diameter is not defined, as for LeechVine, a liana. Boxes of neighbouring stems overlap heavily; masks of the same instances stay separable. A matched plantation scene with the same instance count shows the contrast plainly: boxes stay well separated instead of overlapping, and 9 of 11 detections receive a valid diameter, against several *n/a* here.

Depth improves AP_{50} in all six MaskDINO pairs (Table 3), by 9.68 to 16.93 points. The Mask R-CNN pair moves by only 0.32 points, inside the run-to-run spread of Section 3.4, so that row is best read as showing no measurable effect in either direction; both Mask R-CNN arms remain far below the MaskDINO family. Independent evidence that the pretraining confound is severe comes from Yu et al. (2026), who segment tree trunks from real under-canopy RGB-D imagery with both arms started from COCO-pretrained weights: adding depth moves AP_{50} from 83.9 to only 84.0. Read against the measurement above, that is what a pretrained-on-both-arms comparison is expected to produce.

3.3 Backbone selection

Table 4 compares backbones under a common protocol to select among candidates rather than report converged accuracy: under an identical recipe and seed, FocalNet-L reaches 57.18 at 75 000 iterations and 62.29 at 155 000. The table separates the FocalNet family from the rest by a margin far larger than the spread among the Swin variants or ResNet-50, but does not separate FocalNet-B from FocalNet-L, so both were carried to the full multitask protocol. Swin-L, carried to the same protocol as an audit, matches on diameter but its recall is six to seven points below both FocalNet variants; since detection rather than measurement precision limits how much of a stand the system reports on (Section 3.5), detecting fewer trees is not offset by measuring them slightly better. For segmentation alone, FocalNet-B matches FocalNet-L at roughly half the parameters, and the Mask R-CNN reference confirms that much of the performance belongs to the query-based architecture rather than the backbone.

Table 3: Effect of the depth channel under matched initialisation. Both arms are trained from scratch for 75 000 iterations on identical data with the diameter objective disabled; only the input channel count differs.

Architecture	Backbone	RGB	RGB-D	Δ
Mask R-CNN	ResNet-50	6.54	6.86	+0.32
MaskDINO	ResNet-50	32.00	47.94	+15.93
	Swin-T	33.91	50.84	+16.93
	Swin-B	38.95	48.63	+9.68
	Swin-L	40.75	50.82	+10.07
	FocalNet-B	43.75	58.10	+14.35
	FocalNet-L	44.25	57.18	+12.92

Table 4: Backbone comparison under RGB-D from-scratch training at 75 000 iterations, diameter objective disabled. *The two highest-AP₅₀ backbones, carried forward to the full multitask protocol (Table 2).

Architecture	Backbone	AP ₅₀ (%)	Recall (%)	Params (M)
Mask R-CNN	ResNet-50 (ref.)	6.86	14.06	43.99
MaskDINO	ResNet-50	47.94	45.65	43.82
	Swin-T	50.84	48.73	52.96
	Swin-B	48.63	45.01	113.04
	Swin-L	50.82	46.52	222.80
	FocalNet-B*	58.10	58.02	113.91
	FocalNet-L*	57.18	57.02	233.17

3.4 Diameter loss weighting schedule

Because segmentation is replicated across decoder layers while diameter contributes a single scalar, the diameter term supplies only 1.6 to 2.2% of the total loss at nominally comparable weights. One response is to raise λ_{DBH} once segmentation has formed: this is tested with a pair of runs sharing every setting except the weight schedule, the constant-weight run being the final model itself; the schedule holds $\lambda_{DBH} = 5$ to iteration 60 000, ramps linearly to 50 by 85 000, and holds thereafter.

Table 5: Segmentation loss, excluding the diameter term, for the weighted-schedule run against the constant-weight run at matched iterations.

Iterations	λ_{DBH}	Schedule run	Constant run	Relative gap (%)
30 000–40 000	5	109.35	107.35	+1.87
45 000–55 000	5	103.68	101.58	+2.08
55 000–60 000	5	101.62	99.47	+2.16
62 000–68 000	≈ 14	100.11	97.50	+2.68
70 000–76 000	≈ 28	99.43	95.48	+4.13
80 000–86 000	41 to 50	99.32	93.80	+5.88

The gap grows monotonically, from 1.87 % in the first pre-ramp window to 5.88 % once the schedule completes, against the roughly 2 % variation observed before the schedule diverges. The schedule run plateaus near 99.3 from iteration 62 000 onward while the constant-weight run improves to 93.80, so segmentation does not deteriorate so much as stop improving; the unweighted diameter loss meanwhile shows no consistent benefit. A tenfold increase in the diameter weight cost roughly 5.9 % in segmentation loss and returned no measurable improvement in diameter, consistent with the diameter gradient perturbing shared features rather than with a signal too weak to learn from.

3.5 Error stratification by camera distance

Because depth is an input, every ground-truth instance carries a physical distance computed from geometry alone, so the mixture aggregate scores describe can be decomposed without retraining and without letting the model influence the binning.

Table 6: Detection, classification, and mask quality by camera distance, test split, score threshold 0.30, matching at IoU ≥ 0.5 . Species accuracy and mask IoU are conditional on the instance having been detected.

Distance	N	Recall (%)	Species correct (%)	Mask IoU (%)
0 m to 5 m	1889	82.4	99.6	79.1
5 m to 10 m	1375	79.3	99.7	75.8
10 m to 15 m	744	70.4	99.8	69.8
15 m to 20 m	472	59.5	100.0	65.4
20 m to 30 m	501	46.7	99.6	63.4
> 30 m	267	34.5	98.9	61.0
No valid geometry	2677	42.4	99.2	64.8
All	7925	62.0	99.6	72.2

The three columns behave differently, and that difference is the finding. Recall falls by 58 % in relative terms from the nearest bin to the farthest; mask quality falls by 23 %; species accuracy falls by only 0.7 % and is at or above 98.9 % in every bin. Of 7925 ground-truth instances, 4913 are detected and 4891 of those are also correctly named, so 3034 constitute the total failure, of which 3012 are simply not detected and 22 are detected but misnamed. Misclassification accounts for 0.7 % of all failure, so the species classifier leaves essentially no room for improvement; what fails is detection.

3.5.1 Diameter error by distance

Table 7: Diameter error by camera distance, test split, with the mean ground-truth diameter of each bin. Bracketed intervals are bootstrap 95 % confidence intervals. The geometric baseline of Eq. 2 is shown for reference.

Distance	N	Mean GT (cm)	RMSE (cm)	Rel. RMSE (%)	Geometric (cm)
0 m to 5 m	856	24.56	6.96	28.3	15.22
5 m to 10 m	590	32.22	8.19	25.4	15.44
10 m to 15 m	235	36.80	7.88	21.4	17.60
15 m to 20 m	109	44.26	11.07	25.0	20.33
20 m to 30 m	85	51.60	10.39	20.1	20.36

Absolute RMSE rises with distance (Table 7), which reads as distance degrading the measurement, but the mean ground-truth diameter rises over the same bins from 24.56 to 51.60 cm: distant trees in this data are systematically larger, because small distant stems are the ones that fail to be annotated or to yield a valid geometry. Normalised by that, relative error is approximately flat between 20 and 28 %. The learned head outperforms the geometric baseline in every bin, by roughly a factor of two.

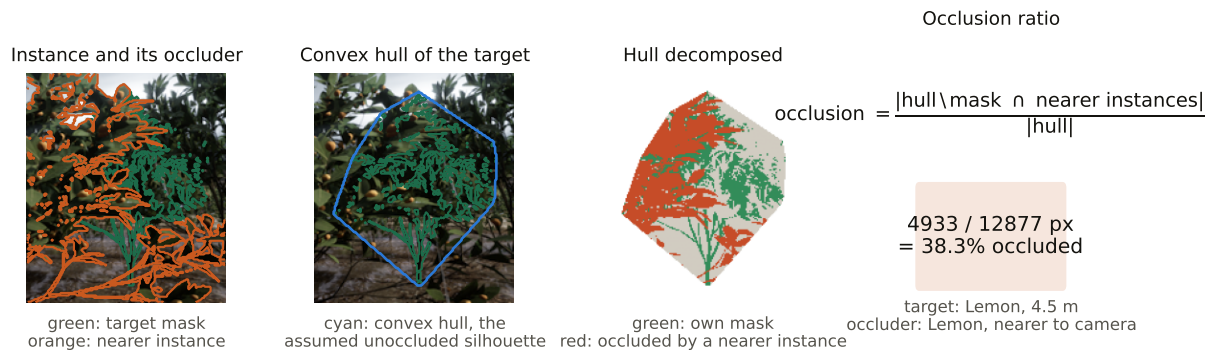


Figure 4: The occlusion ratio traced on one Lemon instance, 4.5 m from the camera and occluded by a nearer Lemon. The convex hull stands in for the unoccluded silhouette, and the share of it covered by nearer instances is the ratio, 38.3 % here.

Distance is a convenient axis but not itself the mechanism, since a distant trunk is harder mainly because more vegetation stands in front of it. Fig. 5 re-bins the same detections by occlusion

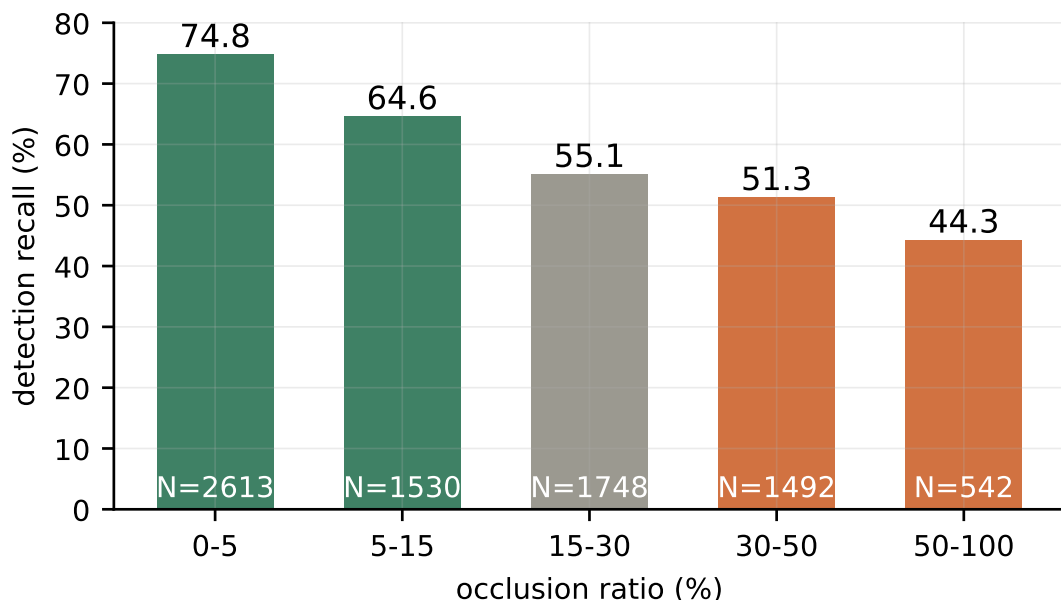


Figure 5: Detection recall against occlusion ratio, the share of an instance’s convex hull occupied by instances nearer to the camera. Recall declines monotonically by 30.5 percentage points from barely occluded to heavily occluded stems.

ratio (defined in Fig. 4): recall declines monotonically by 30.5 percentage points from barely occluded to heavily occluded stems, which places occlusion alongside distance as a direct driver of the detection shortfall rather than a consequence of it.

3.6 Sensor resolution as a floor on diameter accuracy

Table 8: Physical width of one pixel at the native resolution, and therefore the granularity with which any method can express a trunk width at that distance.

Distance	One pixel (cm)
5 m	2.08
10 m	4.17
20 m	8.33
30 m	12.50
20 m at 1920×1080 (4× native)	2.08

At 20 m, the evaluation cap, a one-pixel difference in measured trunk width corresponds to 8.33 cm, the same magnitude as the end-to-end RMSE reported above. A method cannot express a trunk width more finely than the imagery quantises it. Reaching a floor of 2 cm at this range needs a focal length roughly four times the current 240 px; rendering at 1920×1080 rather than 480×270 reaches 2.08 cm. This gives a future SPREAD rendering pass a concrete target.

The prediction pattern in Fig. 6 is consistent with this floor. Where the evidence is coarse, predicting near the conditional mean is the error-minimising choice, so the shrinkage reflects the quantisation the head works against rather than a defect of the head.

4 Discussion

4.1 Diameter accuracy under shared-feature supervision

Diameter accuracy in the multitask setting is bounded by something other than the amount of supervision the diameter head receives; two pieces of evidence, both internal to the multitask runs themselves, indicate interference in the shared features is the limiting factor. First, the

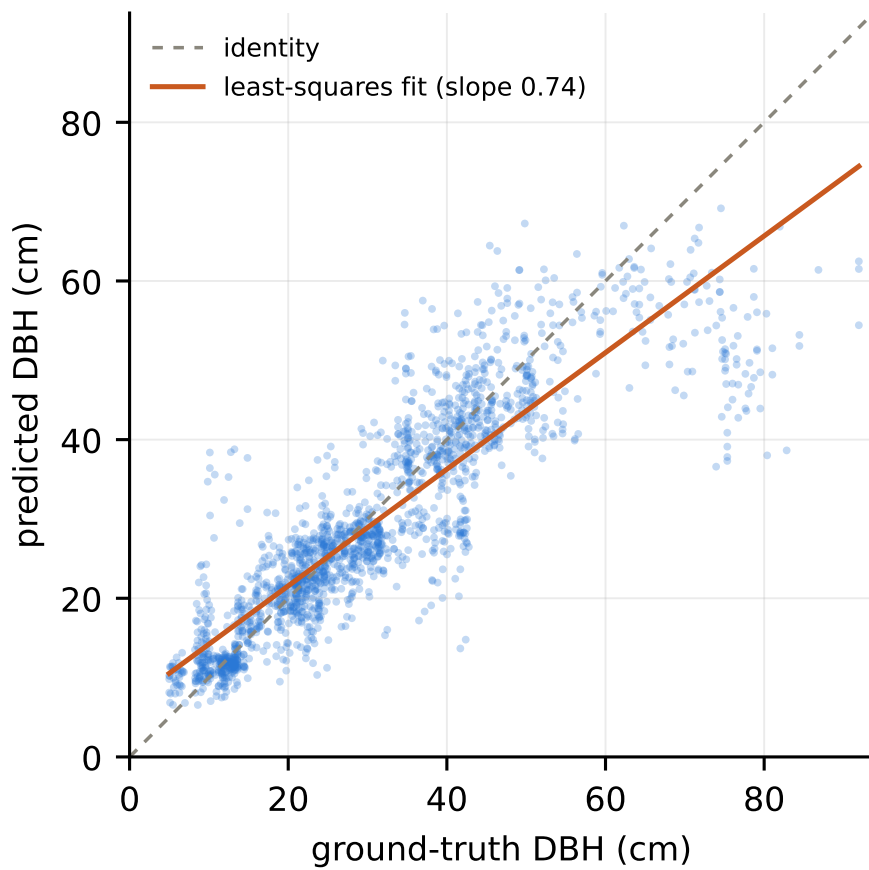


Figure 6: Predicted against ground-truth DBH for the final model. The least-squares slope of 0.74 shows regression towards the mean, small stems over-estimated and large stems under-estimated, the expected response of a regressor whose input cannot separate neighbouring diameters at this resolution.

head is not short of optimisation: in each of four multitask runs trained to 155 000 iterations, validation AP_{50} peaks at the final iteration while the best diameter RMSE arrives between 26 and 55 % of the schedule earlier. Across the evaluation points beyond iteration 50 000, diameter RMSE moves within a band of 0.20 cm standard deviation while AP_{50} over the same span rises by 12.33 percentage points; the correlation between the two series is -0.04 . Second, the two objectives cannot be served by one set of parameters at the same point in training: because the optima are separated by tens of thousands of iterations, checkpoint selection becomes a trade-off rather than a joint optimum. Third, Section 3.4 tests the response that follows most naturally from the hypothesis that the diameter signal is too weak, raising the diameter loss weight, and finds it counterproductive, which is what interference in the shared features predicts and a weak diameter signal does not.

4.2 Implications of the distance stratification

Species classification is highly accurate in this setting, accounts for 0.7 % of total failure, and does not degrade with distance. Mask quality declines gradually. Detection declines sharply and is the main limiting factor of the three. This has a consequence for how such systems should be reported: a single recall figure of 62.0 % is a weighted mixture of 83 % nearby and 32 % far, with the weights set by the distance distribution of the evaluation data rather than by any property of the model, so two studies reporting the same aggregate number on stands of different density are not reporting the same thing. Real imagery corroborates that occlusion, not distance alone, drives the shortfall: Duclos et al. (2026) name tree occlusion, alongside species imbalance, as one of the two most pressing obstacles they encounter.

The same pattern appears when the evaluation population is split by scene: plantation images average 5.31 instances each against 12.47 for rainforest images, and AP_{50} reaches 76.27 % on plantation images against 45.98 % on rainforest images, a gap of 30.29 points. Specialising on

plantation images alone reaches 74.06 % AP_{50} against 69.95 % for the jointly trained model on the same images, a cost of 4.11 points, while on rainforest images specialisation changes the result by only 0.04 points; the rainforest deficit therefore appears to belong to the domain itself, and training both domains together gives up a small, one-sided amount of accuracy rather than trading one domain for the other, which is why the final model is not split into domain specialists. Lowering the score threshold from 0.50 to 0.30 raises recall by 14.8 points in the nearest bin but by 27.9 points at 15 m to 20 m, showing that many distant stems are detected with low confidence rather than missed outright: part of the distance penalty is a calibration effect recoverable by choosing an operating point deliberately, and part is genuine non-detection.

4.3 Relation to prior work

Direct numerical comparison against published diameter errors requires care, because reported error depends on the sensing platform, the stand density, and the distance range admitted. Close-range protocols reach a few centimetres because at one to two metres the pixel floor is roughly 0.5 cm, an order of magnitude finer than this study's 20 m cap, and Wu et al. (2025) photographs single rubber trees at that range with a physical reference marker. Terrestrial laser scanning (Liang et al., 2018) operates at a different level of accuracy altogether, while Grondin et al. (2022), the nearest structural precedent, likewise trains on synthetic renders and reports error growing with camera distance.

Within one study and protocol, Duclos et al. (2026) report AP_{50} of 90.8 without species and 51.2 with them, so requiring species costs 39.6 points on their own data. Against the one row sharing the species-aware task, this study is 11.1 points higher on AP_{50} but lower on the supporting metrics, by 12.4 points of AP and 18.4 of AR_{50} , matching what Section 3.5 measures internally: at a loose overlap threshold the model is accurate on the trees it detects but does not detect enough of them. Resolution is not a limiting factor only in synthetic imagery; Duclos et al. (2026), working on photographed under-canopy scenes, likewise report substantial gains from higher image resolution and judge it likely to be fundamental to progress.

4.4 Limitations and future work

Synthetic training data. All training and evaluation use rendered imagery; no independent real-world evaluation is reported. No public under-canopy dataset shares SPREAD's species composition, so any transfer test available today would confound domain shift with taxonomic shift, leaving the result uninterpretable as evidence about either effect alone. Closing this gap requires either a species-matched real dataset or fine-tuning data collected under the same taxonomic scope as SPREAD.

Detection under occlusion. Explaining the detection shortfall is not closing it. Recall still falls to 34.5 % beyond 30 m, and rainforest segmentation remains far below plantation. Of 6233 eligible test trees, 66.9 % are matched to a detection at all; of those, only 40.5 % yield a usable diameter. A tree that is detected but whose mask cannot resolve its trunk is, for inventory purposes, indistinguishable from one never detected.

Diameter definition and resolution. Diameter is not equally well posed across species. Four of the eleven supervised species are small fruit trees that branch at or below breast height, so a single diameter at 1.3 m is not the quantity a field crew would record for them; they are supervised anyway because excluding them would narrow what the model can report on. This one, unlike the others, is a data generation decision and is directly addressable in future rendering passes.

Future work. Three directions follow. Collecting or annotating a real under-canopy dataset that shares SPREAD's species composition would let a transfer test separate domain shift from taxonomic shift. Rendering SPREAD at higher resolution would raise the pixel floor directly. And an architectural change aimed at detection may be more direct than further refinement of species classification or diameter estimation, since detection accounts for 99.3 % of failure at long range while the other two already perform well on the trees that are detected; amodal completion and aggregating evidence across nearby frames are two candidate directions, though neither is tested here.

This study set out to determine whether a single network can deliver the two measurements forest inventory records, from the viewpoint a field crew occupies, and where such a system actually fails.

Depth is not a marginal input in this setting. Under controls that remove the usual confound with pretrained initialisation, adding a depth channel raises AP_{50} by 12.92 percentage points for FocalNet-L and up to 16.93 points across the six MaskDINO backbones tested. It is also what makes diameter recoverable at all, since converting an apparent trunk width to a physical one requires a distance.

A single multitask network reaches 62.29 % test AP_{50} while producing both outputs in one pass, at no measurable cost to segmentation. Evaluated end-to-end on its own detections, it returns a diameter, 7.62 cm test RMSE, for 27.1 % of eligible trees; the shortfall is split roughly evenly between trees the model never detects and trees it detects but whose predicted mask is too imprecise to resolve a trunk. Diameter accuracy itself is bounded by interference in the shared features rather than by the supervision the head receives, and raising the diameter loss weight degrades segmentation without improving diameter.

Stratifying by camera distance, which the depth input makes possible without retraining, shows that the three abilities the system needs behave very differently. Detection falls from 82.4 % to 34.5 % across the distance range and accounts for 99.3 % of all failure. Species identification of detected trees holds at 98.9 to 100% throughout. Relative diameter error is approximately flat once the systematic increase in tree size with distance is accounted for. The system is limited by detecting stems, not by recognising or measuring them, so the classifier and the regressor leave little room for improvement.

For the exact architecture and training configuration, the student's own code, and the full reference list, see the appendices of the 28-page research paper this document summarises.

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